

Hepatoerythropoietic Porphyrria (HEP)

CLINICAL MANIFESTATIONS

In some patients, liver function test results have been abnormal, but serum iron and iron binding are normal. The clinical manifestations of HEP resemble a combination of those seen in Porphyrria Cutanea Tarda (PCT) and Congenital Erythropoietic Porphyrria (CEP).

LABORATORY FINDINGS

The urinary porphyrin pattern in HEP is similar to that observed in PCT. However, elevated red blood cell (RBC) protoporphyrin (PROTO) levels help distinguish HEP from PCT. Characteristic findings in HEP include:

- Elevated urinary uroporphyrin I and III (URO-COPRO ratio > 5:1)
- Urinary 7-carboxyl porphyrins with more than 90% as isomer III
- Increased fecal coproporphyrin (COPRO) and isocoproporphyrin (ISOCOPRO)
- Elevated RBC PROTO (zinc-chelated)

These findings indicate abnormal porphyrin synthesis in both the liver and the bone marrow. In contrast to PCT, serum iron concentrations are typically normal in HEP.

HISTOPATHOLOGY

Skin biopsy specimens from blistering lesions show subepidermal bullae, similar to those seen in PCT. Periodic acid–Schiff (PAS)-positive material is present in

and around dermal capillaries. However, skin biopsy is not required for diagnosis.

DIFFERENTIAL DIAGNOSIS

Conditions to consider include: - Childhood porphyrias (e.g., homozygous or compound heterozygous variants of Variegate Porphyria [VP] and Hereditary Coproporphyria [HCP]) - Severe forms of PCT - Congenital Erythropoietic Porphyria (CEP)

Unlike HEP, CEP is associated with erythrodontia and mutilating skin scarring.

TREATMENT

There is no specific treatment for HEP other than strict photoprotection to prevent skin damage.

Pseudoporphyria

Pseudoporphyria refers to a condition in which patients exhibit cutaneous symptoms resembling PCT but lack the characteristic porphyrin abnormalities.

ETIOLOGY

It may be induced by certain medications, including: - Furosemide - Nalidixic acid - Tetracycline - Naproxen - Pyridoxine - Isotretinoin

It can also occur in patients with chronic renal failure undergoing long-term hemodialysis. In these cases, plasma porphyrin levels may be moderately elevated (2- to 4-fold) due to impaired renal clearance and inadequate removal by dialysis filters.

HISTOPATHOLOGY

- Blistering is subepidermal with minimal or no dermal inflammation
- PAS staining shows little or no deposition around upper dermal blood vessels
- Indirect immunofluorescence on split skin reveals no dermal deposition of fluorescent material
- Direct immunofluorescence may show patchy granular deposition of IgG and C3 at the basement membrane zone — a finding also seen in PCT and epidermolysis bullosa acquisita

Erythropoietic (EPP)

Protoporphyria

AT A GLANCE

- Second most common cutaneous porphyria
- Autosomal dominant inheritance with deficiency of ferrochelatase
- Onset typically in early childhood (1–4 years); late onset is rare
- Cutaneous manifestations: erythema, edema, purpura, skin thickening, waxy scars; blistering is uncommon
- Symptoms: acute burning and stinging of the skin, often severe
- Approximately 5% of patients may develop potentially fatal hepatic failure